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Project 1: Gale-Shapley Algorithm Variants

Another variation of the Gale-Shapley algorithm could involve forming groups of three instead of two. In this scenario, there would be twice as many people in the second group than in the first group. Each person in the first group would be able to propose to anyone in the second group, just like in the normal Gale-Shapley, until the first set of pairs were formed. Then each person in the first group would be able to propose again to anyone in the second group. If the proposee was already paired in the first round, they would still be able to compare and break their pairing if desired. However, the person in the first group who is now unmatched would immediately propose to their first partner before the second round continues. This would continue until everyone in the first group is matched with two people from the second group.

To implement this algorithm, you would need two queues, one for the people in the first group without a partner, and one for people in the first group with one partner already. Any time the first queue was empty, the next person from that queue would propose to someone in the second group like usual. If the first queue was empty, the algorithm would move to the second queue. The person in the first group would then make a proposal to another person in the second group, possibly breaking a pairing between two or three people in the process. This would continue until both queues are empty and all people from the first group are paired with two people from the second group.

This problem could come up in similar situations to many of the applications of the usual Gale-Shapley algorithm (besides monogamous, heterosexual relationships, of course). Employers looking for multiple candidates, teams looking to draft multiple players, or schools looking for multiple students are all possibilities. Of course, in all these scenarios, groups much larger than three are also possible, and the algorithm would likely follow a similar situation. Variations of this problem include not allowing the second round of picks to break pairings made in the first round, allowing anyone to propose to any two people (taking the average ranking of a pair when proposing to a third person), having blocks between people in the first and second group or between people within the second group, and allowing ties.